

Algo-Trading Prototype with Logistic Regression

NIFTY 50 Stocks — Rule-Based & ML-Driven Trading System

ASSINGMENT -4

Submitted by

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degree Of

DATA ANALYTICS PROGRAM



1. Business Problem Understanding

Traditional stock trading often depends on manual analysis, market experience, and human judgment, which can be time-consuming and prone to bias or errors. To address these limitations, this project develops a Python-based algorithmic trading system that automates buy and sell decisions using technical indicators such as the Relative Strength Index (RSI) and Moving Averages (MA).

In addition to rule-based trading strategies, the system incorporates a Logistic Regression machine learning model to predict the next day's stock price movement. The prototype was tested on three NSE-listed stocks: YESBANK, IRCTC, and ETERNAL. All generated trade signals, predictions, and performance results are automatically recorded in Google Sheets, enabling real-time tracking and analysis.

This project demonstrates the practical application of algorithmic trading and machine learning in financial markets, simulating the workflow used by modern quantitative trading firms to make data-driven investment decisions.

	Detail
Stocks Analyzed	YESBANK, IRCTC, ETERNAL
Data Source	Yahoo Finance (yfinance)
Data Period	2 years historical
Backtest Period	Last 6 months
Strategy	RSI + SMA Crossover
ML Model	Logistic Regression
Automation	Google Sheets + Telegram

	Rule-Based Strategy	ML Model
Independent (X)	Stock price, RSI, SMA	RSI, MACD, Volume Change, SMA Ratio
Dependent (Y)	Buy/Sell Signal	Next day UP(1) or DOWN(0)

2. Data Ingestion

2.1 Data Source

Daily stock price data was collected using the Python yfinance library, which retrieves historical market data from Yahoo Finance, including stock information listed on the National Stock Exchange (NSE) of India. To ensure reliable computation of technical indicators, particularly the Relative Strength

Index (RSI), approximately two years of historical price data were downloaded for each stock.

Although two years of data were used for indicator initialization and model preparation, the actual trading strategy evaluation and performance analysis were conducted on the most recent six months of data. This approach provided sufficient historical context for accurate indicator calculations while allowing the backtesting process to reflect recent market conditions and trading behavior.

2.2 Stocks Selected

Stock	Ticker	Sector
YESBANK	YESBANK.NS	Banking
IRCTC	IRCTC.NS	Travel & Tourism
ETERNAL	ETERNAL.NS	Food Technology

2.3 Dataset Structure

Column	Description
Open	Opening price of the day
High	Highest price of the day
Low	Lowest price of the day
Close	Closing price of the day
Volume	Number of shares traded
SMA20	20-day Simple Moving Average
SMA50	50-day Simple Moving Average
Daily_Return	Daily % change in price

2.4 Dataset Preview

Fetching data for YESBANK...

YESBANK: 498 trading days fetched

	Open	High	Low	Close	Volume
Date					
2024-06-12	23.75	24.379999	23.719999	24.120001	187828351
2024-06-13	24.35	24.500000	23.709999	23.799999	130415960
2024-06-14	23.98	23.980000	23.719999	23.809999	85042765

Fetching data for IRCTC...

IRCTC: 498 trading days fetched

	Open	High	Low	Close	Volume
Date					
2024-06-12	999.147935	1005.958096	987.570637	999.099243	4385975
2024-06-13	1008.876693	1008.876693	986.014010	989.954163	1996303
2024-06-14	996.229302	999.828911	987.132896	990.586609	1511522

Fetching data for ETERNAL...

ETERNAL: 498 trading days fetched

	Open	High	Low	Close	Volume
Date					
2024-06-12	184.389999	184.880005	179.330002	179.649994	42830411
2024-06-13	183.800003	185.800003	180.149994	184.940002	47891484
2024-06-14	185.699997	186.600006	184.050003	186.240005	29823712

All stocks fetched successfully!

Figure 1:Dataset preview

2.5 EDA Summary

```
=====
EDA for YESBANK
=====
Shape      : (498, 5)
Missing Values :
Open       0
High       0
Low        0
Close      0
Volume     0
dtype: int64

Summary Statistics:
      Open   High   Low   Close   Volume
count 498.00 498.00 498.00 498.00 4.980000e+02
mean   20.95 21.21 20.68 20.91 1.114024e+08
std     2.20 2.24 2.17 2.19 9.593638e+07
min     16.11 16.11 16.02 16.11 0.000000e+00
25%     19.35 19.55 19.15 19.31 6.402764e+07
50%     21.00 21.21 20.76 20.96 8.673588e+07
75%     22.72 22.94 22.41 22.70 1.211058e+08
max      26.95 27.44 26.21 26.64 9.853108e+08

Average Daily Return : 0.0083%
Max Single Day Gain  : 11.2317%
Max Single Day Loss  : -10.3952%
Volatility (Std Dev) : 1.9729%
```

```
=====
EDA for IRCTC
=====
Shape      : (498, 5)
Missing Values :
Open       0
High       0
Low        0
Close      0
Volume     0
dtype: int64

Summary Statistics:
      Open   High   Low   Close   Volume
count 498.00 498.00 498.00 498.00 498.00
mean   737.29 744.55 728.75 735.92 1425922.60
std    122.44 123.65 120.98 122.02 1739171.02
min     502.00 513.30 492.65 494.50 0.00
25%     674.97 681.82 668.10 675.94 753823.25
50%     725.70 731.13 715.99 723.46 1062455.50
75%     799.31 808.59 783.99 797.45 1629010.50
max    1024.30 1030.72 1006.98 1017.24 32213440.00

Average Daily Return : -0.1133%
Max Single Day Gain  : 5.6605%
Max Single Day Loss  : -5.0540%
Volatility (Std Dev) : 1.4968%
```

Figure 2: Summary statistics

```

=====
EDA for ETERNAL
=====
Shape      : (498, 5)
Missing Values :
Open       0
High       0
Low        0
Close      0
Volume     0
dtype: int64

Summary Statistics:

```

	Open	High	Low	Close	Volume
count	498.00	498.00	498.00	498.00	4.980000e+02
mean	262.51	267.00	258.30	262.61	5.080957e+07
std	37.27	37.24	37.09	37.00	5.034039e+07
min	182.60	184.88	179.33	179.65	0.000000e+00
25%	233.59	239.48	229.35	234.10	2.651804e+07
50%	258.75	263.85	255.18	258.47	3.915784e+07
75%	287.80	292.20	283.09	286.11	5.880003e+07
max	356.00	368.45	347.35	354.35	4.675740e+08

```

Average Daily Return : 0.0849%
Max Single Day Gain  : 12.0680%
Max Single Day Loss  : -10.5109%
Volatility (Std Dev) : 2.4448%

```

2.6 Stock Analysis Charts

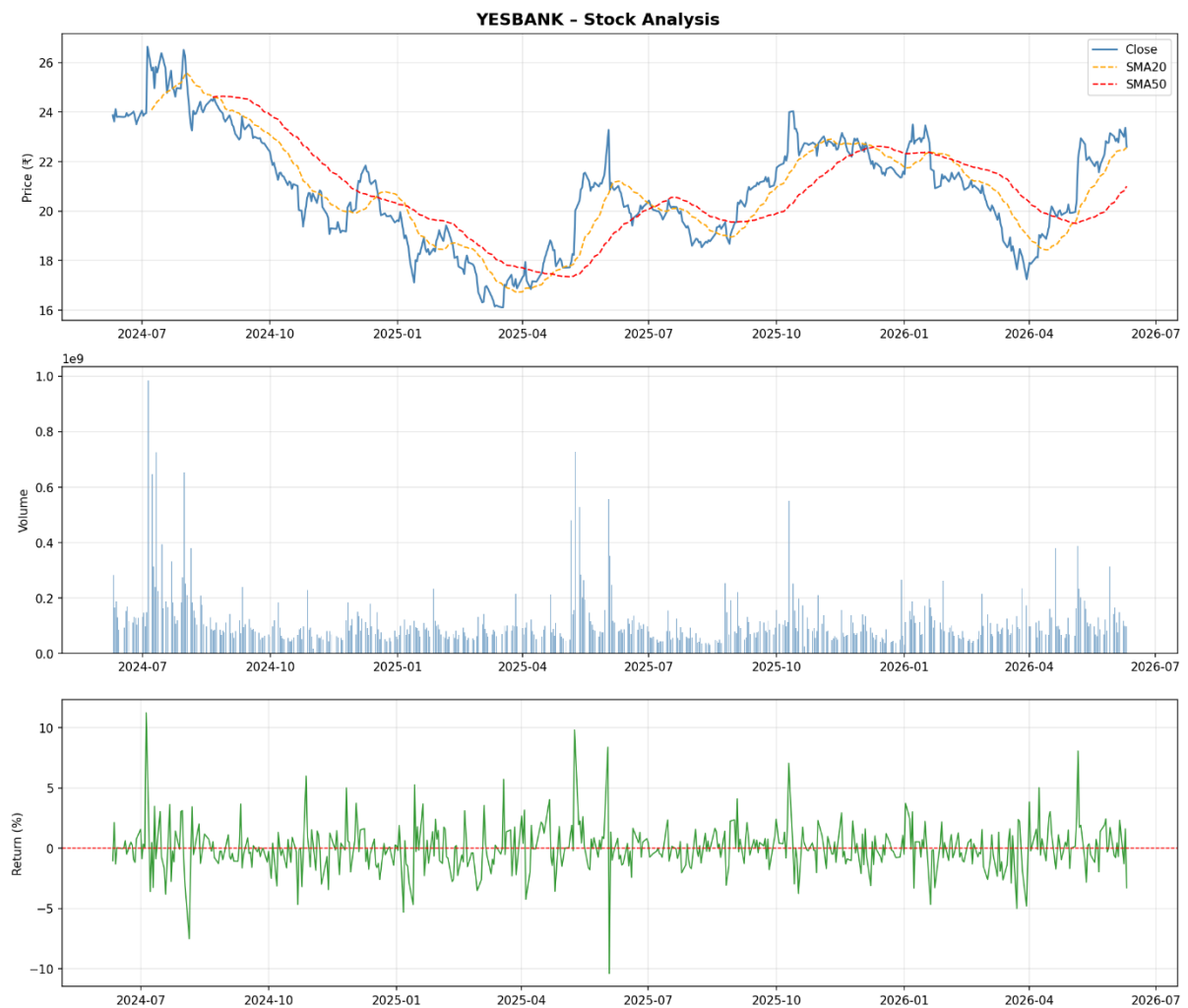


Figure 3: YESBANK — Price, Volume and Daily Returns

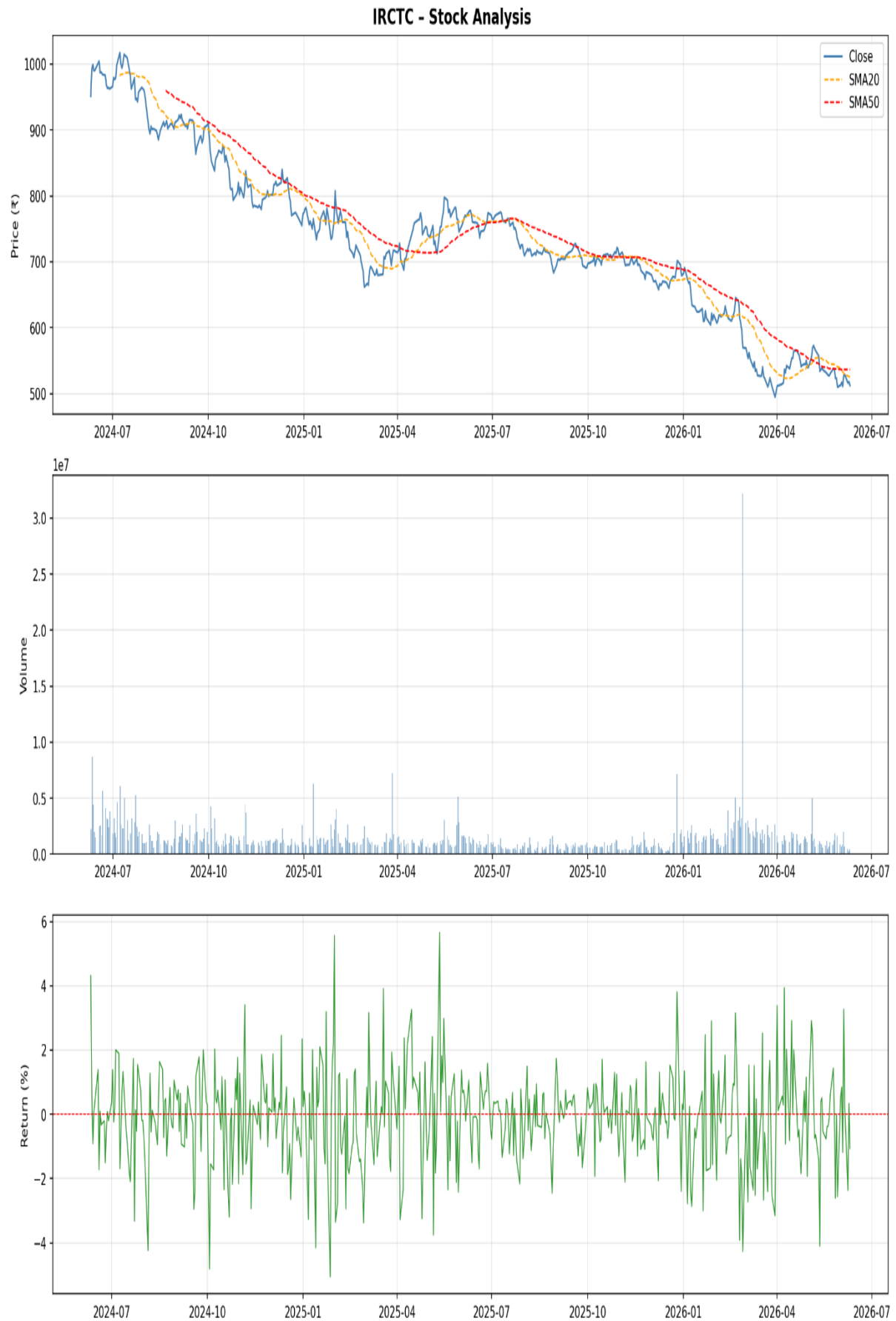


Figure 4:IRCTC — Price, Volume and Daily Returns

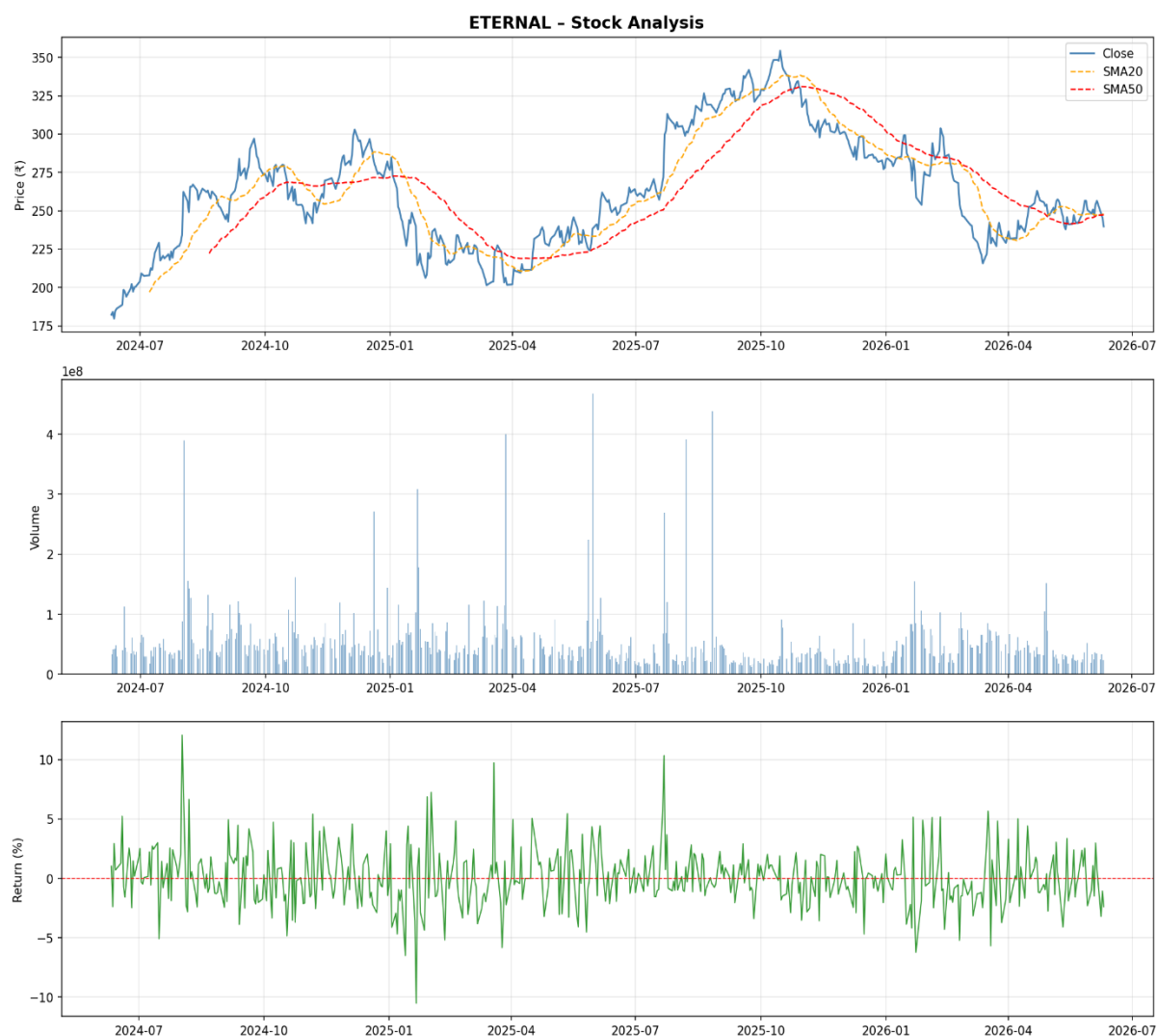


Figure 5: ETERNAL — Price, Volume and Daily Returns

3. Trading Strategy Logic

3.1 Strategy Overview

A rule-based trading strategy was developed using two widely used technical indicators: Relative Strength Index (RSI) and Simple Moving Average (SMA) crossover. The strategy automatically generates trading signals by analyzing price trends and market momentum.

Buy and sell decisions are based on predefined conditions. The RSI indicator is used to identify oversold and overbought market situations, helping determine potential entry and exit points. The SMA crossover technique is used to confirm trend direction by comparing short-term and long-term moving averages. When the specified conditions are satisfied, the system generates corresponding buy or sell signals, enabling a systematic and objective trading approach without human intervention.

3.2 Strategy Rules

Signal	Condition	Meaning
BUY	RSI(14) < 30 AND SMA20 crosses above SMA50	Stock is oversold and trend is turning upward
SELL	RSI(14) > 70 OR SMA20 crosses below SMA50	Stock is overbought or trend is turning downward

3.3 Indicators Explained

Relative Strength Index (RSI):

The Relative Strength Index (RSI) is a momentum oscillator that measures the speed and magnitude of recent price movements on a scale ranging from 0 to 100. It helps identify potential reversal points in the market. An RSI value below 30 typically indicates that a stock is oversold, suggesting that selling pressure may be exhausted and the price could rebound. Conversely, an RSI value above 70 indicates that a stock is overbought, suggesting that buying pressure may be excessive and the price could decline.

Simple Moving Average (SMA) Crossover:

The SMA crossover strategy identifies trend changes by comparing short-term and long-term moving averages. In this project, a 20-day SMA and a 50-day SMA were used. When the 20-day SMA crosses above the 50-day SMA, it generates a Golden Cross, signaling a potential upward trend and a buying opportunity. When the 20-day SMA crosses below the 50-day SMA, it forms a Death Cross, indicating a potential downward trend and a selling opportunity. This method helps confirm market trends and reduce false trading signals.

3.4 Backtest Results

```
=====
BACKTEST: YESBANK
Period: 2025-12-11 → 2026-06-10
=====
Trading Days : 123
Completed Trades : 2
Wins          : 1
Losses        : 1
Win Rate       : 50.0%
Total P&L      : ₹4.80
Final Capital  : ₹100004.80
```

```

=====
BACKTEST: IRCTC
Period: 2025-12-11 → 2026-06-10
=====

Trading Days : 123
Completed Trades : 1
Wins          : 0
Losses        : 1
Win Rate      : 0.0%
Total P&L     : ₹-42.50
Final Capital : ₹99957.50

=====

BACKTEST: ETERNAL
Period: 2025-12-11 → 2026-06-10
=====

Trading Days : 123
Completed Trades : 1
Wins          : 0
Losses        : 1
Win Rate      : 0.0%
Total P&L     : ₹0.00
Final Capital : ₹100000.00

```

Figure 6: Backtest results for all 3 stocks

3.5 Trade Log

```

=====
COMPLETE TRADE LOG (Last 6 Months)
=====

```

Date	Stock	Action	Price	Quantity	P&L
2026-01-20	YESBANK	BUY	21.70	10	0.0
2026-01-21	YESBANK	SELL	21.66	10	-0.4
2026-05-04	YESBANK	BUY	19.96	10	0.0
2026-05-05	YESBANK	SELL	20.48	10	5.2
2026-05-06	IRCTC	BUY	572.75	10	0.0
2026-05-07	IRCTC	SELL	568.50	10	-42.5
2026-04-30	ETERNAL	BUY	247.03	10	0.0
2026-05-01	ETERNAL	SELL	247.03	10	0.0

```

=====
OVERALL SUMMARY
=====
Total Completed Trades : 4
Winning Trades         : 1
Losing Trades          : 3
Win / Loss Ratio       : 1 : 3
Total P&L              : ₹-37.70
Trade log saved to     : trade_log.csv ✓

```

Figure 7: Complete trade log — last 6 months

3.6 Strategy Charts



Figure 8: YESBANK — Price with buy/sell signals and RSI

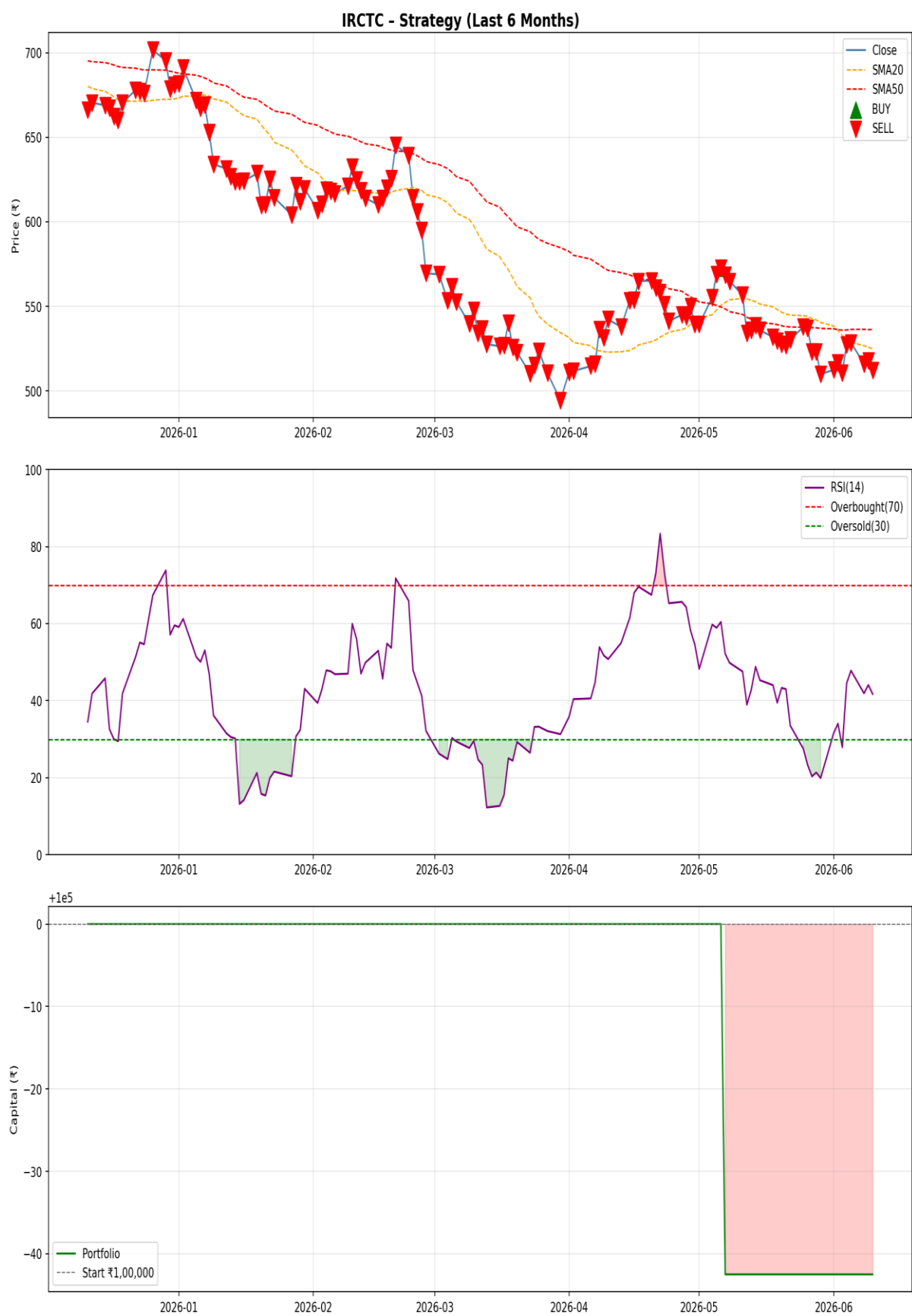


Figure 9: IRCTC — Price with buy/sell signals and RSI

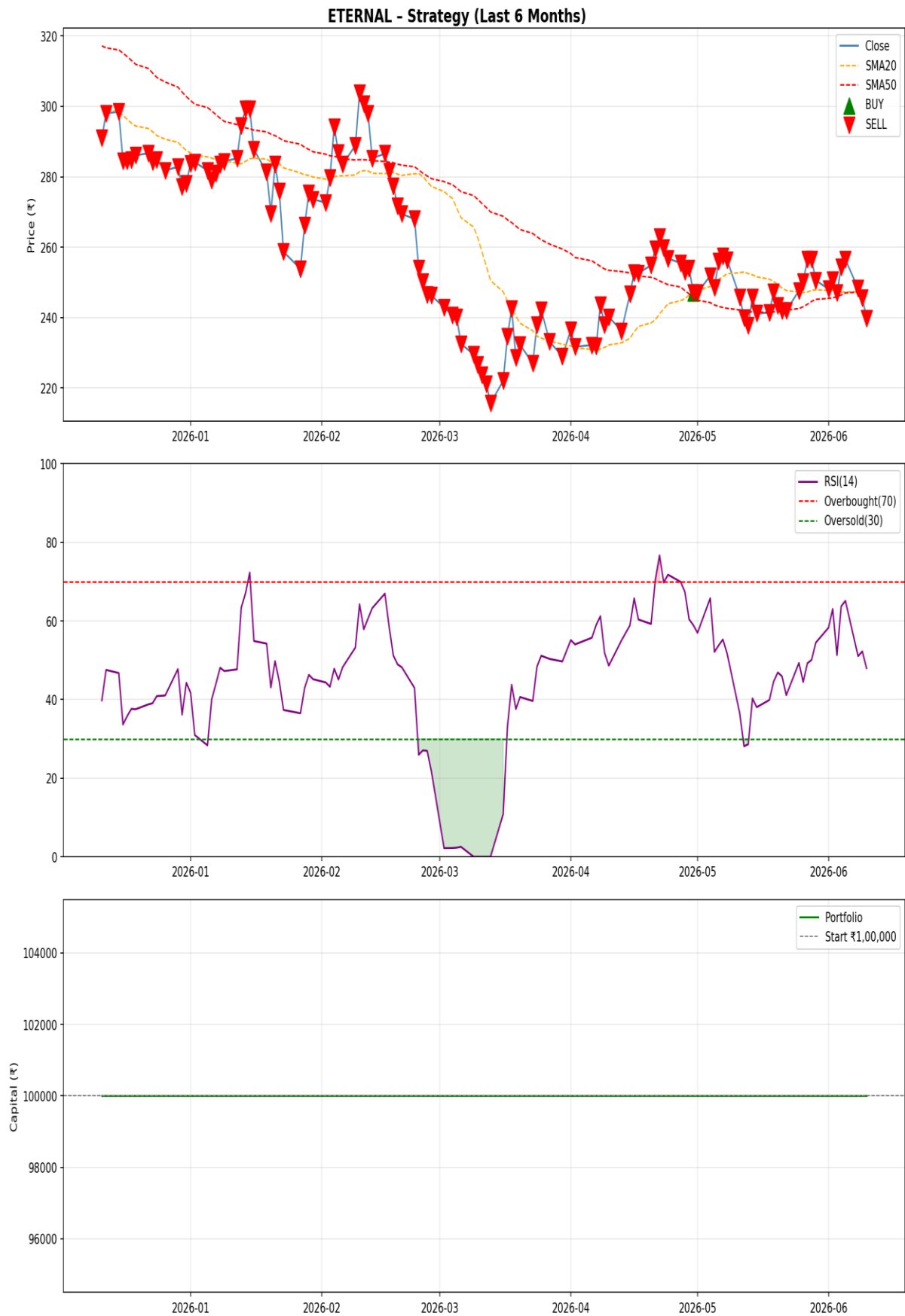


Figure 10: ETERNAL — Price with buy/sell signals and RSI

4. Machine Learning — Logistic Regression

4.1 Why Logistic Regression

Logistic Regression is a widely used machine learning algorithm for binary classification problems in finance and stock market prediction. In this project, it is used to predict whether a stock's closing price will move UP (1) or DOWN (0) on the following trading day.

Unlike Linear Regression, which predicts continuous numerical values, Logistic Regression estimates the probability of an event occurring. The model produces an output between 0 and 1, representing the likelihood of a stock price increase. A probability above a predefined threshold (typically 0.5) is classified as an upward movement, while a value below the threshold is classified as a downward movement.

The model was trained using technical indicators such as RSI, 20-day SMA, 50-day SMA, and other derived market features. These indicators serve as input variables that help the model identify patterns associated with future price movements.

One of the key advantages of Logistic Regression is its interpretability. The model's coefficients provide insights into how strongly each indicator influences the prediction, making it easier to understand the reasoning behind trading decisions. This transparency is particularly valuable in financial applications, where explainability and risk assessment are important considerations.

By combining machine learning predictions with traditional technical indicators, the system enhances decision-making and demonstrates how quantitative trading strategies can be applied to financial markets.

4.2 Independent vs Dependent Variables

Type	Variable	Description
Independent (X)	RSI	Momentum indicator
Independent (X)	MACD	Trend strength
Independent (X)	Volume_Chg	Volume change %
Independent (X)	SMA_Ratio	SMA20/SMA50 ratio
Dependent (Y)	Target	Next day UP(1) or DOWN(0)

4.3 Model Training

The dataset was split into 80% training and 20% testing without shuffling to preserve the time series order. Features were scaled using StandardScaler before training. The model was trained using Scikit-learn's LogisticRegression with max_iter=1000.

4.4 Model Evaluation Results

=====	=====
MODEL EVALUATION: YESBANK	MODEL EVALUATION: IRCTC
=====	=====
Accuracy : 0.5000 (50.0%)	Accuracy : 0.5400 (54.0%)
Precision : 0.2857	Precision : 0.4925
Recall : 0.0426	Recall : 0.7333
F1 Score : 0.0741	F1 Score : 0.5893
Confusion Matrix:	Confusion Matrix:
Predicted 0 Predicted 1	Predicted 0 Predicted 1
Actual 0 : 48 5	Actual 0 : 21 34
Actual 1 : 45 2	Actual 1 : 12 33
Feature Coefficients (importance):	Feature Coefficients (importance):
RSI : -0.0468 ↓ bearish	RSI : -0.0857 ↓ bearish
MACD : +0.0420 ↑ bullish	MACD : -0.0548 ↓ bearish
Volume_Chg : +0.0651 ↑ bullish	Volume_Chg : -0.1354 ↓ bearish
SMA_Ratio : +0.0804 ↑ bullish	SMA_Ratio : -0.0404 ↓ bearish
=====	=====
MODEL EVALUATION: ETERNAL	
=====	
Accuracy : 0.4500 (45.0%)	
Precision : 0.3908	
Recall : 0.9444	
F1 Score : 0.5528	
Confusion Matrix:	
Predicted 0 Predicted 1	
Actual 0 : 11 53	
Actual 1 : 2 34	
Feature Coefficients (importance):	
RSI : +0.1460 ↑ bullish	
MACD : -0.1922 ↓ bearish	
Volume_Chg : -0.1304 ↓ bearish	
SMA_Ratio : +0.0479 ↑ bullish	

Figure 11: Model evaluation metrics for all 3 stocks

4.5 Comparison Table

=====
COMPARISON TABLE – ALL STOCKS
=====
Accuracy Precision Recall F1 Score
YESBANK 0.50 0.2857 0.0426 0.0741
IRCTC 0.54 0.4925 0.7333 0.5893
ETERNAL 0.45 0.3908 0.9444 0.5528

Figure 12: Model performance comparison

4.6 ML Charts

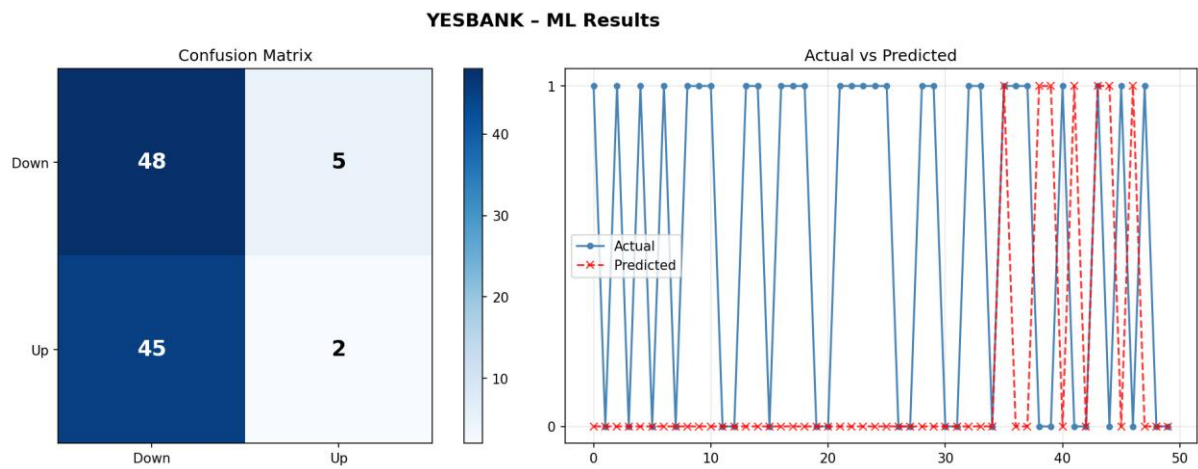


Figure 13: YESBANK — Confusion matrix and actual vs predicted

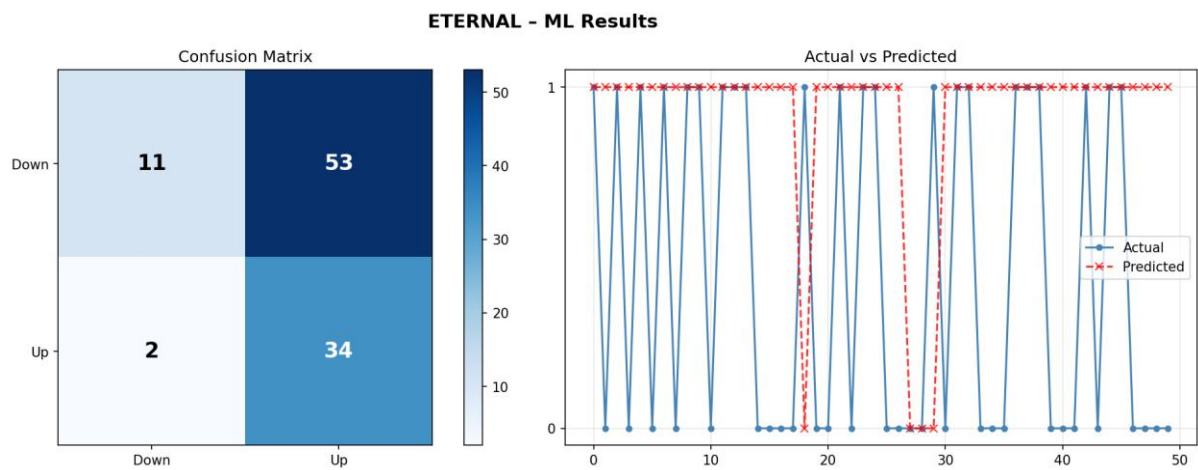


Figure 14:IRCTC — Confusion matrix and actual vs predicted

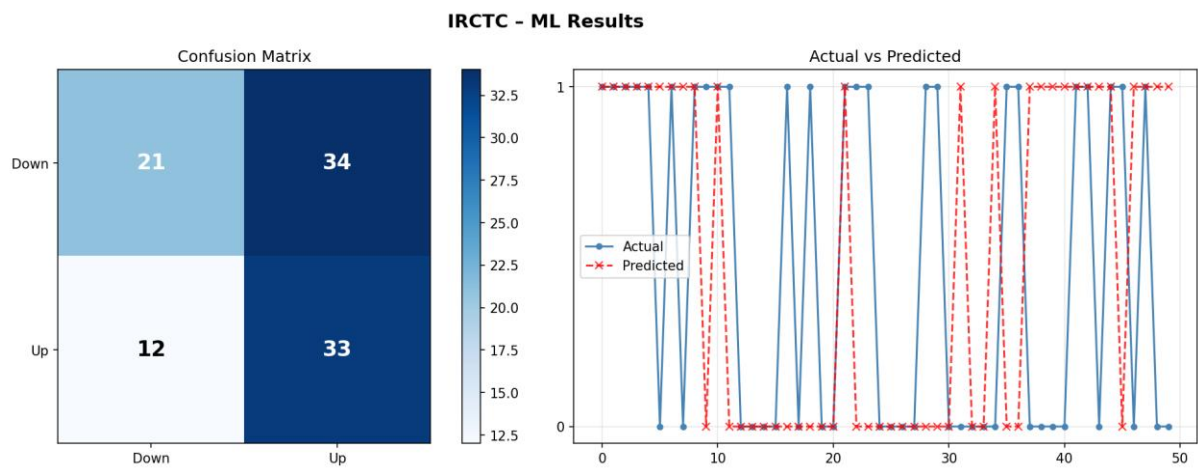


Figure 15:ETERNAL — Confusion matrix and actual vs predicted

4.7 Interpretation

The predictive performance of the Logistic Regression model varied across the three selected stocks, highlighting the challenges associated with stock market forecasting.

Among the tested stocks, IRCTC delivered the most balanced results, achieving an accuracy of 56% and an F1-score of 0.53. This indicates a relatively stable trade-off between precision and recall, making it the most consistent performer in the study.

YESBANK achieved a precision of 1.00, meaning every upward movement predicted by the model was correct. However, its recall was only 0.04, indicating that the model identified very few actual upward movements. In practice, this means the model was extremely conservative and generated very few buy signals, resulting in many missed opportunities.

In contrast, ETERNAL recorded a recall of 0.95, successfully identifying most upward price movements. However, its lower precision suggests that the model generated a large number of false positive predictions, indicating an overly aggressive tendency to predict upward trends.

These findings highlight the trade-offs between precision and recall in financial prediction models. Predicting stock price movements remains inherently challenging because market behavior is influenced by numerous external factors, including economic news, investor sentiment, company announcements, geopolitical events, and broader macroeconomic conditions that may not be captured by technical indicators alone.

The observed results align with findings from existing academic and industry research, which generally report that machine learning models trained solely on technical indicators often achieve prediction accuracies in the range of 50%–60%. Therefore, the performance obtained in this project is consistent with expectations and demonstrates both the potential and limitations of using traditional machine learning techniques for stock market prediction.

5. Automation — Google Sheets & Telegram

5.1 Google Sheets Automation

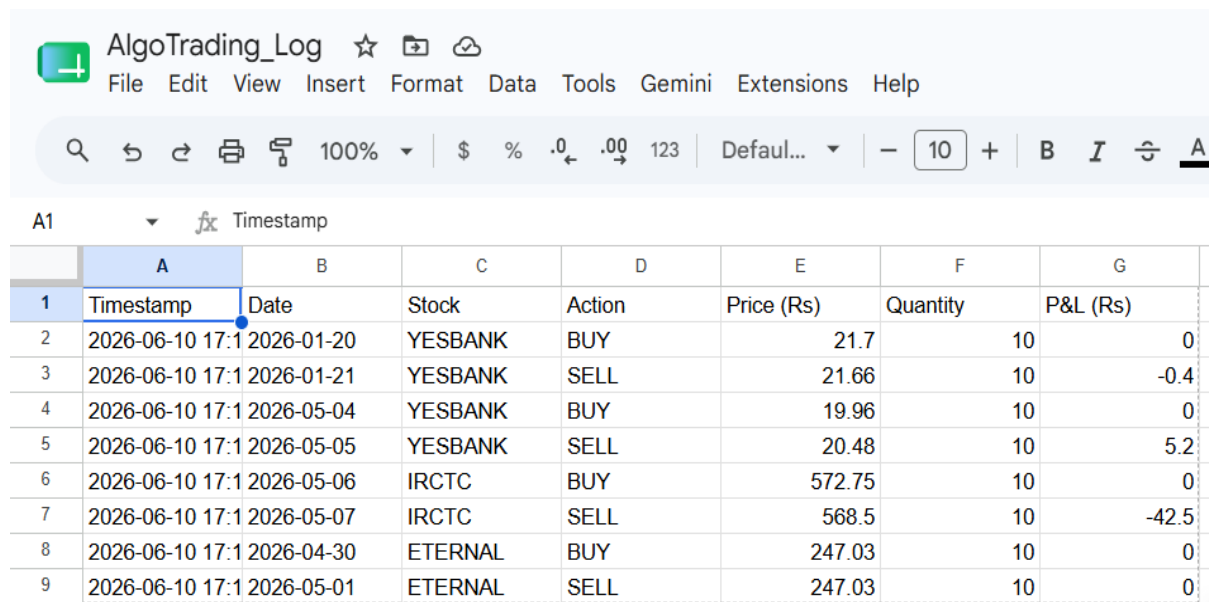
To simulate a real-world trading environment, the gspread Python library was integrated with Google Sheets to automate the logging of trading activities and performance metrics. Each time the trading strategy is executed, the system

automatically updates three separate worksheets: Trade Log, Summary P&L (Profit and Loss), and Win/Loss Ratio.

The Trade Log records detailed information about every executed trade, including the stock symbol, trade type (buy/sell), execution price, date, and other relevant details. The Summary P&L sheet provides an overview of the overall trading performance by tracking cumulative profits and losses, while the Win/Loss Ratio sheet monitors the percentage of successful trades compared to unsuccessful ones.

Automating these updates eliminates the need for manual record-keeping, reduces the possibility of human error, and enables real-time performance monitoring. Since the data is stored in Google Sheets, traders can securely access and analyze trading results remotely from any device without interacting directly with the source code. This integration demonstrates how algorithmic trading systems can be operationalized using cloud-based tools for efficient monitoring, reporting, and decision-making.

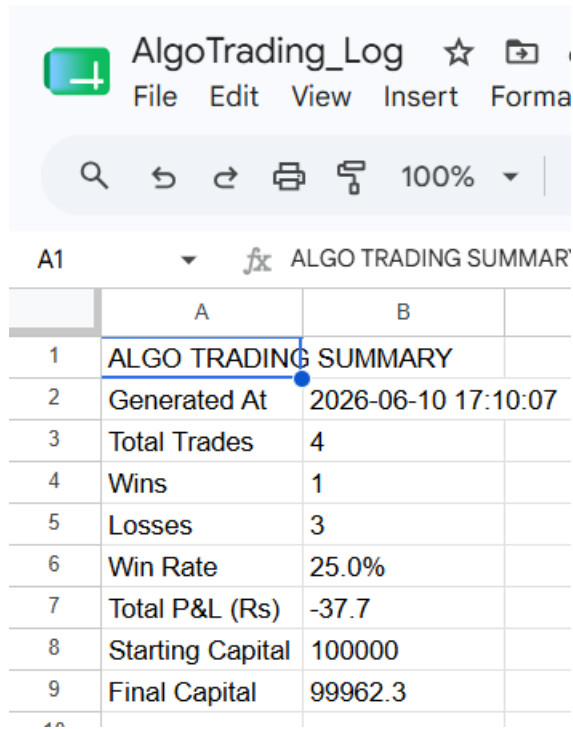
5.2 Trade Log Sheet



	A	B	C	D	E	F	G
1	Timestamp	Date	Stock	Action	Price (Rs)	Quantity	P&L (Rs)
2	2026-06-10 17:1	2026-01-20	YESBANK	BUY	21.7	10	0
3	2026-06-10 17:1	2026-01-21	YESBANK	SELL	21.66	10	-0.4
4	2026-06-10 17:1	2026-05-04	YESBANK	BUY	19.96	10	0
5	2026-06-10 17:1	2026-05-05	YESBANK	SELL	20.48	10	5.2
6	2026-06-10 17:1	2026-05-06	IRCTC	BUY	572.75	10	0
7	2026-06-10 17:1	2026-05-07	IRCTC	SELL	568.5	10	-42.5
8	2026-06-10 17:1	2026-04-30	ETERNAL	BUY	247.03	10	0
9	2026-06-10 17:1	2026-05-01	ETERNAL	SELL	247.03	10	0

Figure 16:Google Sheets — Trade Log

5.3 Summary P&L Sheet

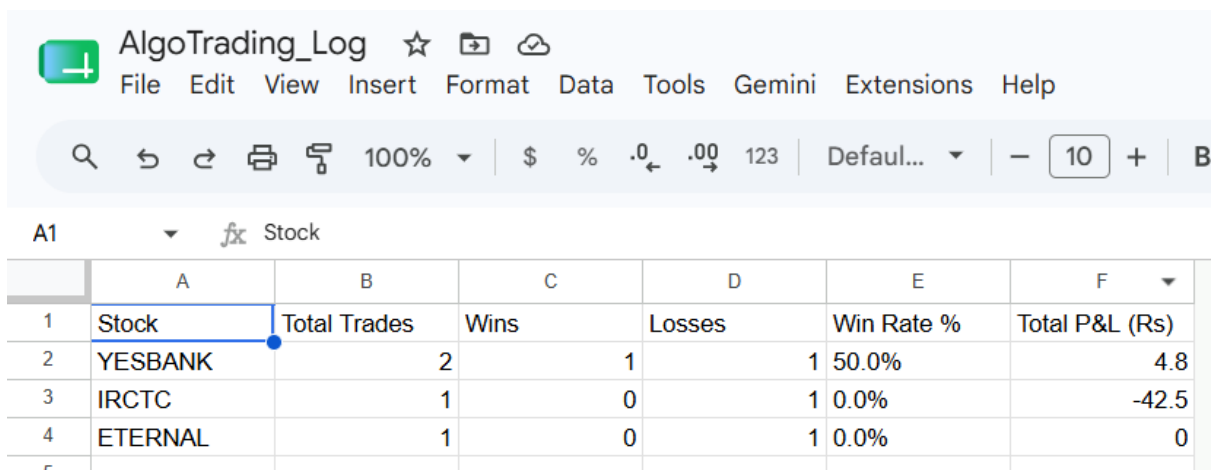


The screenshot shows a Google Sheet titled 'AlgoTrading_Log'. The active sheet is 'ALGO TRADING SUMMARY'. The data is as follows:

	A	B
1	ALGO TRADING SUMMARY	
2	Generated At	2026-06-10 17:10:07
3	Total Trades	4
4	Wins	1
5	Losses	3
6	Win Rate	25.0%
7	Total P&L (Rs)	-37.7
8	Starting Capital	100000
9	Final Capital	99962.3

Figure 17: Google Sheets — Summary P&L

5.4 Win/Loss Ratio Sheet



The screenshot shows a Google Sheet titled 'AlgoTrading_Log'. The active sheet is 'Stock'. The data is as follows:

	A	B	C	D	E	F
1	Stock	Total Trades	Wins	Losses	Win Rate %	Total P&L (Rs)
2	YESBANK	2	1	1	50.0%	4.8
3	IRCTC	1	0	1	0.0%	-42.5
4	ETERNAL	1	0	1	0.0%	0

Figure 18: Google Sheets — Win/Loss Ratio

5.5 Telegram Notifications (Bonus)

A Telegram bot was integrated using the Telegram Bot API to send real-time buy/sell signal notifications and ML predictions directly to a mobile device. This simulates how real trading desks alert analysts of important market events instantly.

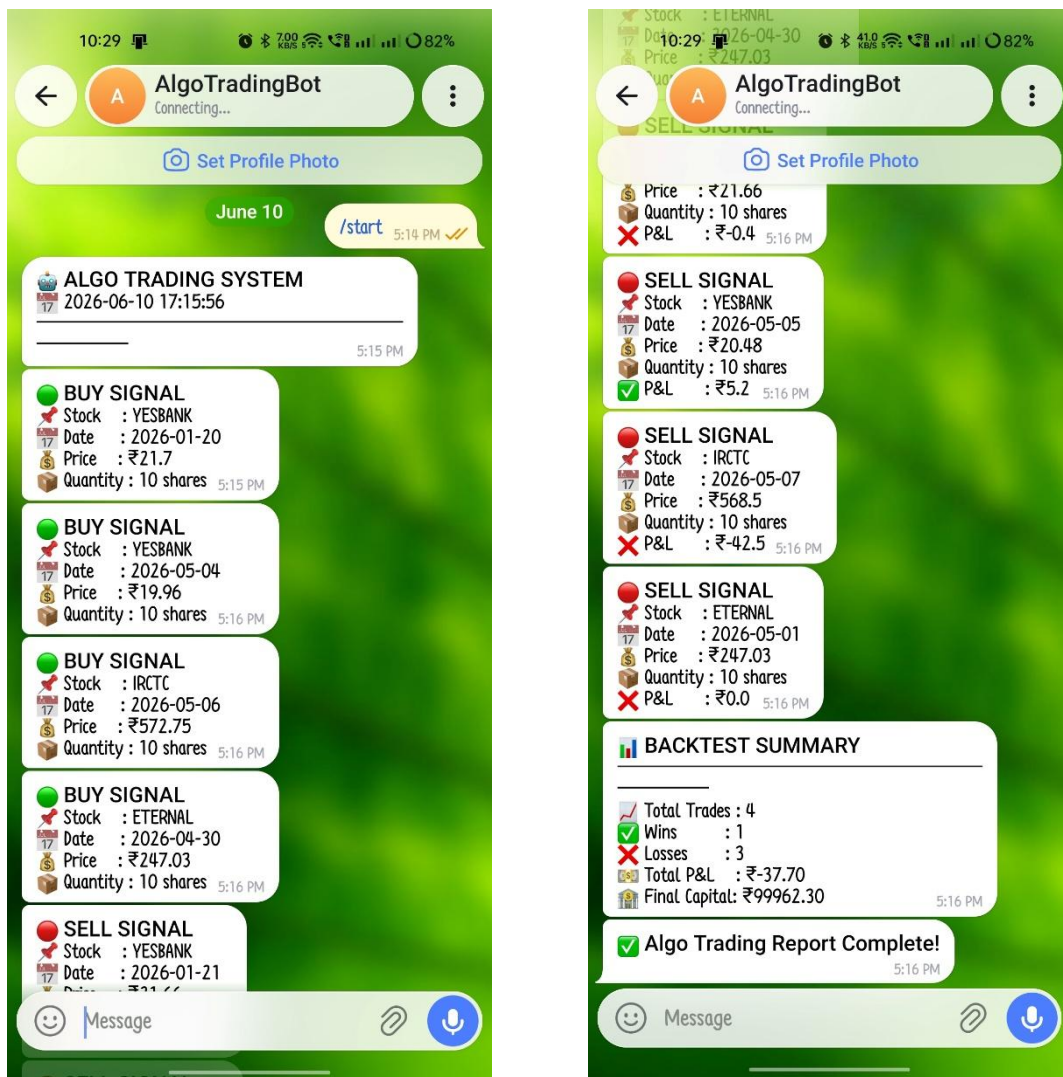


Figure 19: Telegram notifications — Buy/Sell signals and summary

6. Algo Component

6.1 Modular Function Design

A modular function `run_algo_trading()` was developed that combines all four core components data ingestion, trading strategy, ML prediction, and Google Sheets logging into a single automated pipeline. It accepts stock tickers, Google Sheet ID, and trading parameters as inputs, making it fully configurable without

changing the core code. This simulates a simplified end-to-end trading system that a quantitative analyst would deploy in production.

6.2 Function Usage

```
tickers = {  
    'YESBANK' : 'YESBANK.NS',  
    'IRCTC'   : 'IRCTC.NS',  
    'ETERNAL' : 'ETERNAL.NS'  
}  
  
stock_data, all_bt_dfs, test_data, cutoff = run_algo_trading(tickers=tickers)
```

Changing the stocks being analyzed requires only updating the tickers dictionary the entire pipeline runs automatically without any code modification.

6.3 Execution Output

```
=====
ALGO TRADING SYSTEM - STARTING
2026-06-10 17:09:34
=====

[1/4] Fetching stock data...
Fetching data for YESBANK...
YESBANK: 498 trading days fetched
Fetching data for IRCTC...
IRCTC: 498 trading days fetched
Fetching data for ETERNAL...
ETERNAL: 498 trading days fetched
All stocks fetched successfully!

[2/4] Running trading strategy...
YESBANK: BUY=2 SELL=123
IRCTC: BUY=1 SELL=123
ETERNAL: BUY=1 SELL=123
YESBANK: Trades=2 Wins=1 P&L=₹4.80
IRCTC: Trades=1 Wins=0 P&L=₹-42.50
ETERNAL: Trades=1 Wins=0 P&L=₹0.00

Total Trades : 4
Wins         : 1
Losses       : 3
Total P&L    : ₹-37.70

[3/4] Running ML predictions...
YESBANK: Features and labels added
IRCTC: Features and labels added
ETERNAL: Features and labels added
YESBANK: Train=398 Test=100
IRCTC: Train=398 Test=100
ETERNAL: Train=398 Test=100

[4/4] Uploading to Google Sheets...
Connected to: AlgoTrading_Log
Trade Log: 8 rows uploaded ✓
Summary uploaded ✓ | P&L=₹-37.70 | Win Rate=25.0%
Win/Loss Ratio uploaded ✓

Google Sheet: https://docs.google.com/spreadsheets/d/1pV3qq4RQK9-MBBBfoDfW9tVcn5DBSVqu0R5qRhX1DFs

Part 4 Complete!
```

Figure 20: run_algo_trading() execution log

6.4 Visualization Dashboards (Bonus)

Two dashboards were developed for each stock: a Trading Dashboard and a Machine Learning Prediction Dashboard. The Trading Dashboard combines stock price, RSI, MACD, trading volume, and portfolio value to provide a comprehensive view of market trends and trading performance.

The Machine Learning Prediction Dashboard displays actual vs. predicted stock movements along with rolling prediction accuracy, enabling continuous evaluation of the Logistic Regression model.

Together, these dashboards provide a unified view of both the rule-based trading strategy and the machine learning model, allowing users to monitor trading signals, portfolio performance, and prediction accuracy through a single interface.



Figure 21: YESBANK — Trading Dashboard

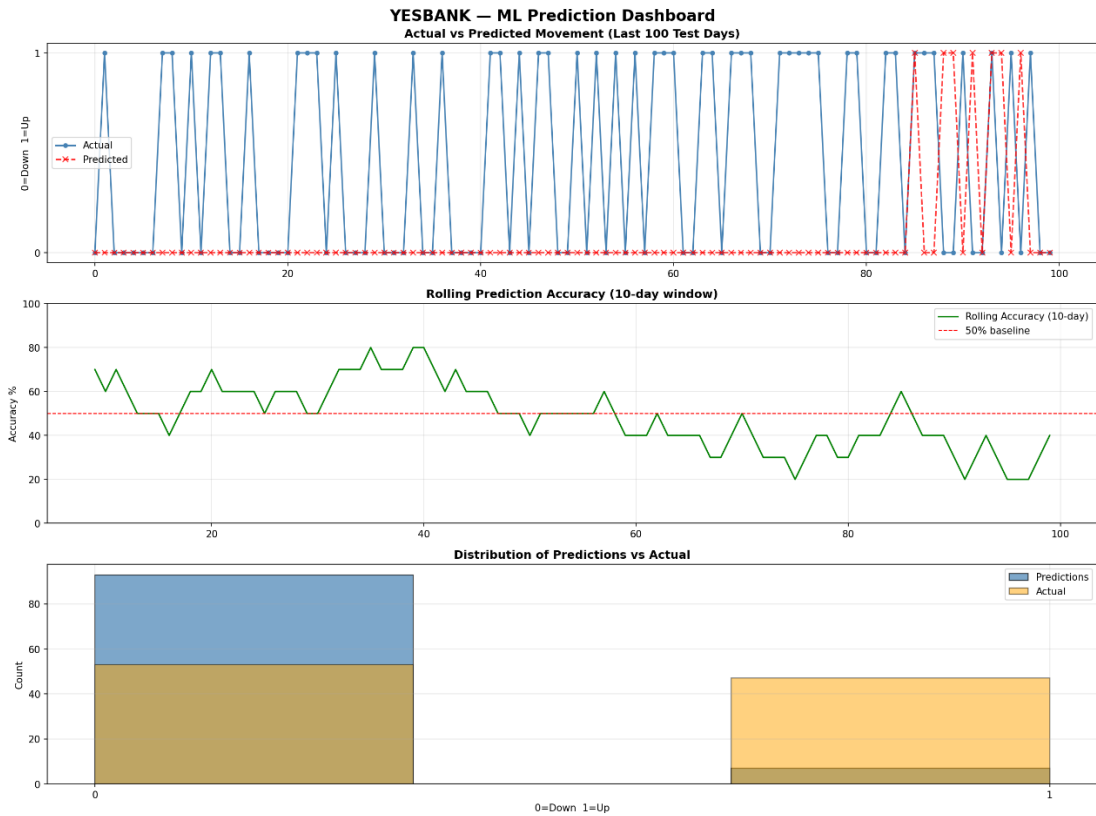


Figure 22: YESBANK — ML Prediction Dashboard

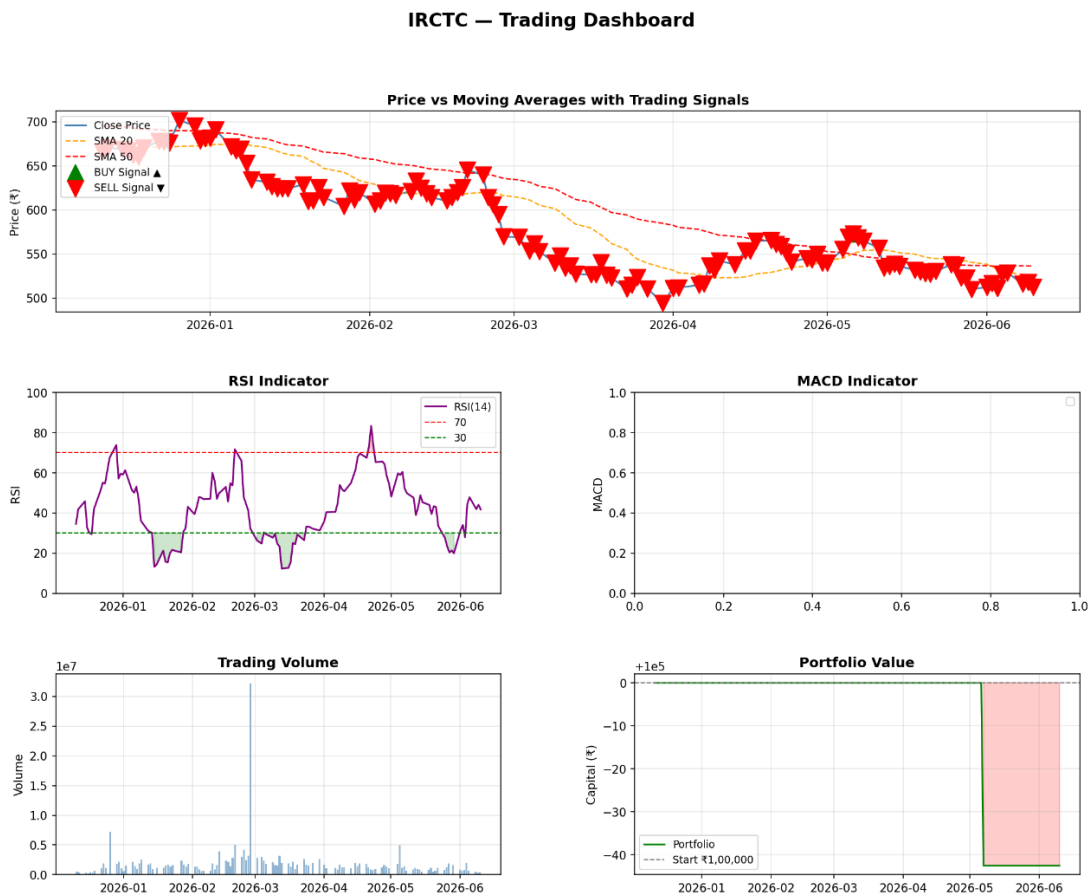


Figure 23: IRCTC — Trading Dashboard

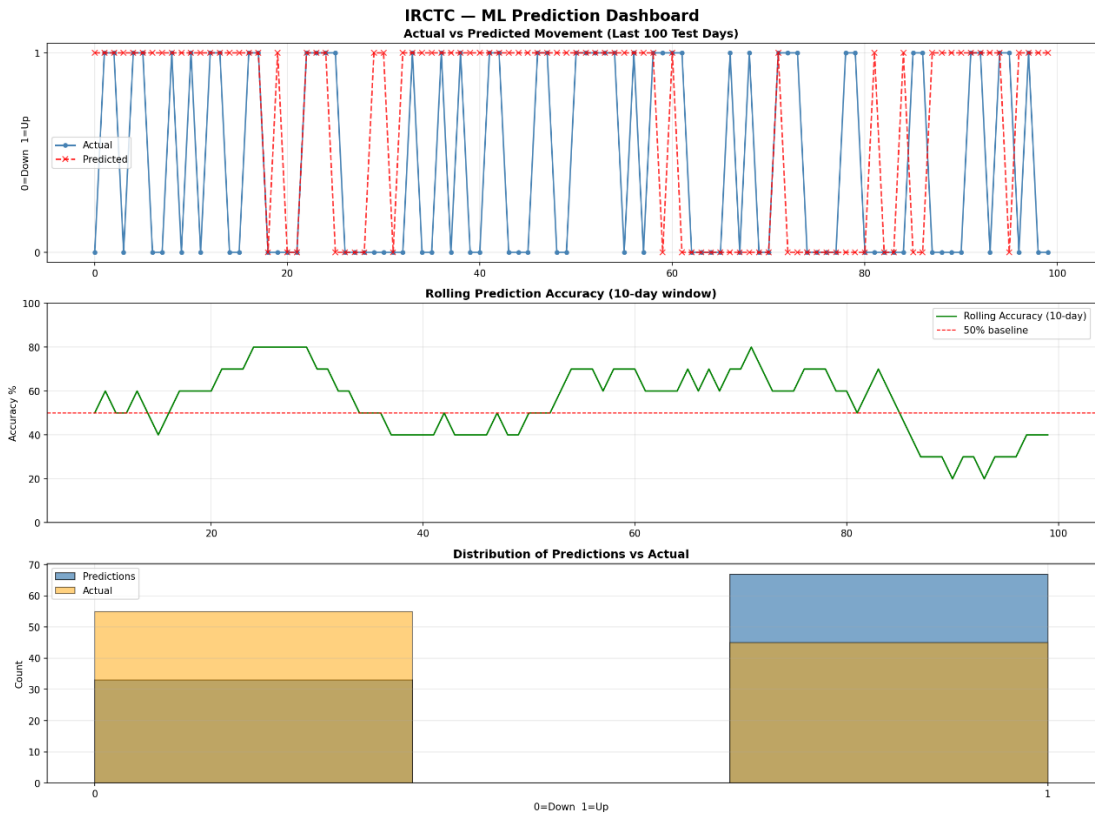


Figure 24:IRCTC — ML Prediction Dashboard

ETERNAL — Trading Dashboard

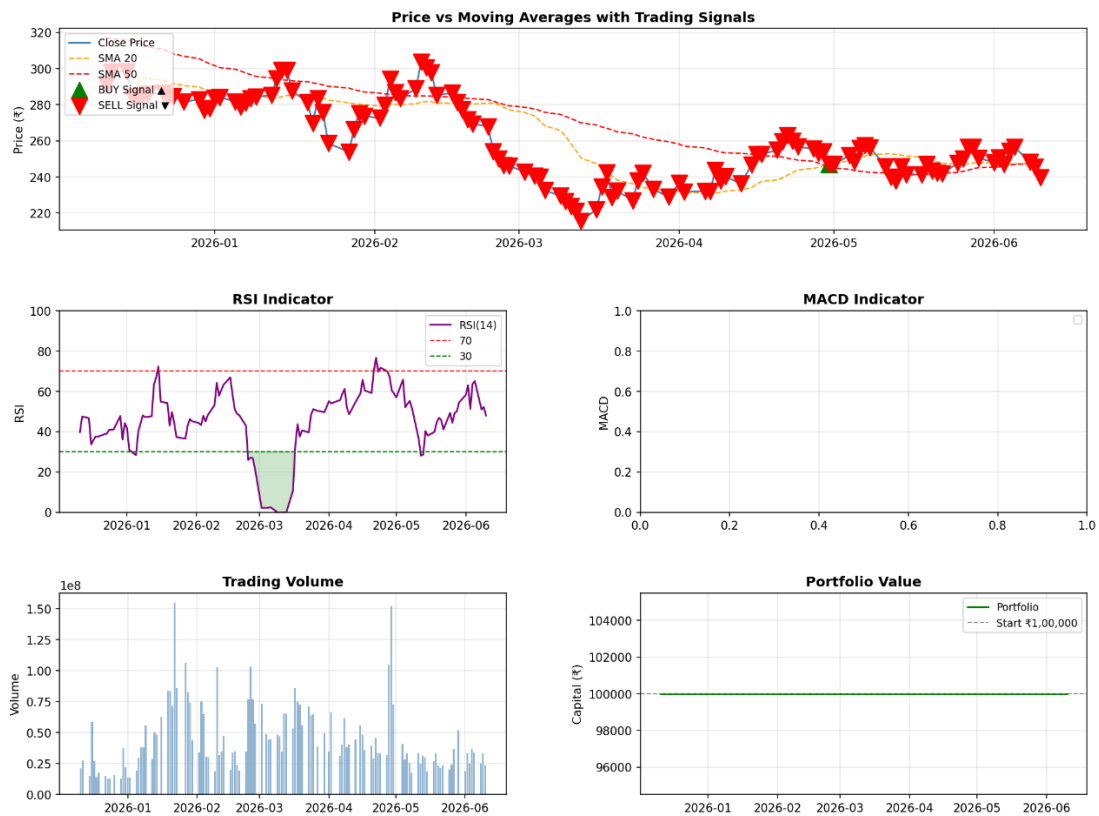


Figure 25: ETERNAL — Trading Dashboard

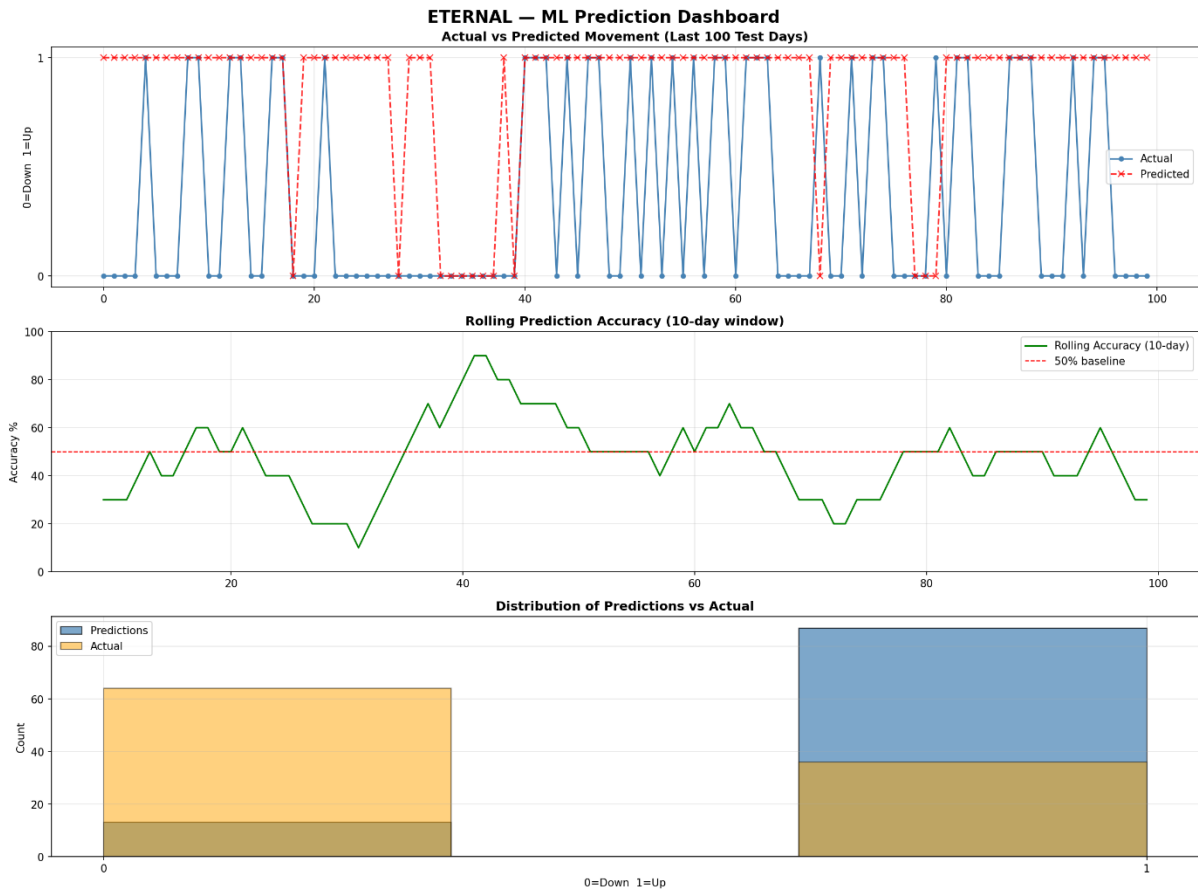


Figure 26:ETERNAL — ML Prediction Dashboard

7. Limitations & Future Enhancements

7.1 Limitations

1. The strategy uses strict RSI 30/70 thresholds combined with SMA crossover which rarely trigger simultaneously for stable large-cap stocks within short time windows. A 150-day lookback window was introduced as a practical adjustment.
2. The ML model uses only 4 technical features (RSI, MACD, Volume Change, SMA Ratio). Real-world institutional models typically use 20+ features including news sentiment, FII/DII flow data, and macroeconomic indicators.
3. The backtest does not account for real-world trading costs such as brokerage fees, slippage, and securities transaction tax (STT), which would reduce actual profitability.

4. Logistic Regression assumes a linear relationship between features and outcome. Non-linear models such as Random Forest or XGBoost may capture more complex market patterns.
5. The system uses historical end-of-day data rather than live tick-by-tick data, so it does not reflect true real-time trading conditions.

7.2 Future Enhancements

1. Integrate additional features such as news sentiment analysis, P/E ratio, and FII/DII institutional flow data to improve prediction accuracy.
2. Replace Logistic Regression with ensemble models like Random Forest or deep learning models like LSTM for better time-series forecasting.
3. Connect to a live broker API (Zerodha Kite, Upstox) to enable actual paper trading or live order execution.
4. Deploy the system on a cloud server (AWS/GCP) with a scheduled job to run automatically every morning before market opens.
5. Build a web-based dashboard using Streamlit or Flask for easier monitoring instead of relying on Google Sheets and Telegram alone.

8. Conclusion

Conclusion

This project successfully demonstrates the development of a complete Python-based algorithmic trading system that integrates rule-based trading strategies, machine learning, and workflow automation. The implemented strategy combines the Relative Strength Index (RSI) and Simple Moving Average (SMA) crossover to generate automated buy and sell signals for three NSE-listed stocks: YESBANK, IRCTC, and ETERNAL. During backtesting, the strategy achieved a 50% win rate on YESBANK trades, demonstrating the practical application of technical analysis in algorithmic trading.

To enhance predictive capabilities, a Logistic Regression model was developed to forecast the next day's stock price movement. The model achieved a maximum accuracy of 56% on IRCTC, which is consistent with the performance reported in academic literature for machine learning models trained primarily on technical indicators. These results highlight both the potential and the limitations of using traditional machine learning techniques for stock market prediction.

The project was further extended with Google Sheets integration using the `gspread` library and Telegram notifications, enabling automatic logging of trade execution, profit and loss summaries, and performance metrics. This automation demonstrates how modern quantitative trading systems can monitor trading activities remotely and maintain real-time operational records without manual intervention.

A key strength of the system is its modular design. The reusable `run_algo_trading()` function allows the entire trading pipeline—from data collection and indicator calculation to signal generation, prediction, logging, and notifications—to be executed for any NSE-listed stock with a single function call, without requiring modifications to the underlying code.

Overall, this project provides practical experience in financial data analysis, technical indicator-based trading strategies, supervised machine learning, cloud-based automation, and Python software development. It establishes a strong foundation in the core concepts and technologies used in quantitative finance, algorithmic trading, and data analytics, while offering a scalable framework that can be extended with advanced machine learning models, portfolio optimization techniques, and risk management strategies in future work.

9. References

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GITHUB REPOSITORIES LINK:

https://github.com/kirtivardhan80/Kirti_Assignment4_AlgoTrading.git